

Solve the Farmer's problem – to get $x_1=60$ and $x_2=20$

```
A=[1 2; 1 3; 1 1];  
b=[100 120 110];  
f=[50 120]; % we are to maximize. Linprog minimizes  
lb=[0 0];  
ub=[110 110];  
  
opts = optimoptions(@linprog,'Display','off','Algorithm','dual-simplex');  
[solution,profit,exitflag]=linprog(-f',A,b',[],[],lb',ub',opts);
```

Solve the problem we massaged

```
A=[4 3 -3 2 1; 1 0 0 -1 2; -1 0 0 1 -2; -1 -1 1 -1 -1];  
b=[10 2 -2 -1];  
-f=[-1 -2 2 -3 -4]; % we are to maximize. Linprog  
minimizes  
lb=[0 0 0 0 0];  
ub=[inf inf inf inf inf];
```

Original problem is unbound (exitflag=-3)

Put an upper bound on the variables, e.g. 10

```
ub=[10 10 10 10 10];
```

Max of 43.3 is obtained for $x_1 = 0.0$, $u_2 = 4.7$, $v_2 = 10$, $x_3 = 10$, $x_4 = 6$, i.e. $x_2 = 4.7 - 10 = -5.3$

Optimization of sale

		pants	jackets	available
cotton	1		1.5	750
polyester	2		1	1000

Objective:

Maximize $f = 50x_1 + 40x_2$

```
A=[1 1.5; 2 1];
```

```
b=[750 1000];
```

```
-f=[-50 -40]; % we are to maximize.
```

Constraints:

$x_1 + 1.5x_2 \leq 750$

$2x_1 + x_2 \leq 1000$

and

$x_1 \geq 0, x_2 \geq 0,$

```
Linprog minimizes
```

```
lb=[0 0];
```

```
ub=[inf inf];
```

Max of 28750 \$ is obtained for $x_1 = 375.0$ and $x_2 = 250$

Message

$$\max f = 3x_1 + 2x_2 - x_3$$

$$x_1 + 2x_2 + x_3 \leq 5$$

$$2x_1 + 4x_2 - x_3 \geq 1$$

$$x_1 \geq 0, x_2 \leq 0$$

Solution

$$\max f = 3x_1 - 2y_2 - u_3 + v_3$$

$$x_1 - 2y_2 + u_3 - v_3 \leq 5$$

$$-2x_1 + 4y_2 + u_3 - v_3 \leq -1$$

$$x_1 \geq 0, y_2 \geq 0, u_3 \geq 0, v_3 \geq 0$$

unbounded

$$x_2 = -y_2$$

$$x_3 = u_3 - v_3$$